# Archit Uniyal

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## Skills

**Programming**:
*Proficient*: Python, C++, C *Familiar*: Java, Javascript
**Frameworks**: Pytorch, Keras, Tensorflow

## Research interests

Differential Privacy, Privacy-preserving ML, Natural Language Processing, Pruning Algorithms

## Publications

Memorization in NLP Fine-tuning Methods
*The 2022 Conference on Empirical Methods in Natural Language Processing*

Quantifying Privacy Risks of Masked Language Models Using Membership Inference Attacks
*The 2022 Conference on Empirical Methods in Natural Language Processing*

DP-SGD vs PATE: Which Has Less Disparate Impact on Model Accuracy?
*ML4Data Workshop at ICML 2021*

## Work Experience

**Research Intern at Samsung Research Institute, Noida** *January 2022 – July 2022*

Conducted research on improving the existing human activity recognition algorithms in wearable trackers.
Conducted research on integrating GAN-based models for better quality of videos in samsung smartphones.

**Researcher at Openmined** *November 2020 – Present*
Conducted research on the disparate impact of DP-SGD and PATE on minority groups in the dataset.
Conducted research on wide neural networks and how they are able to perform better than deep neural networks under specific conditions like differential privacy.

**Research Intern at Accelerating Visions , Dehradun, Uttarakhand** *May 2020 – July 2020*
Built an image search engine powered by deep learning.
Built a custom architecture using image captioning network to perform image search.
Built a Differentially Private image captioning architecture.

**Research Intern at Design Innovation Centre, Panjab University** *June 2019 – July 2019*
Mentors: **Dr. Mamta Juneja (Panjab University)**
Built models for image classification using transfer learning.
Proposed a custom architecture for image classification.
Proposed an algorithm to calculate DDLS and ISNT of a fundus image using image segmentation.

## Projects

**Memorization in NLP Fine-tuning Methods**
This project focuses on the memorization in large language models due to fine-tuning.
We analyze different fine-tuning methods and observe that fine-tuning the head of the models makes it most vulnerable to attacks.

**Quantifying Privacy Risks of Masked Language Models Using Membership Inference Attacks**
This project puts forward an efficient membership inference attack on masked language models (MLM) based on the likelihood ratio hypothesis testing that involves an additional reference MLM to more accurately quantify the privacy risks of memorization in MLMs.

**PrivateClassImbalance**
Conducted experiments on DP-SGD and PATE for epsilon values 0.5, 5 and 15 on MNIST and SVHN datasets.
Conducted an ablation study for determining the optimum number of teachers in PATE to get better performance in terms of fairness and privacy.

**DeepvsWideNN**
Conducted experiments on deep neural networks and wide neural networks to determine how wide networks can perform better with less parameters.
Conducted study on how differential privacy affects their performance.

**Indian Sign Language**
Built an interface to detect alphabets of Indian Sign Language and convert it into text.
InceptionResnetV3 was used as our base architecture to classify the alphabets.
LSTMs were used to generate a sequence of text, which is then fed to the google text to speech API to generate an audio.

## Education

**University Institute of Engineering and Technology, Panjab University, Chandigarh**
B.E in Computer Science August 2018 – July 2022
*CGPA: 8.65 (on a scale of 10)*